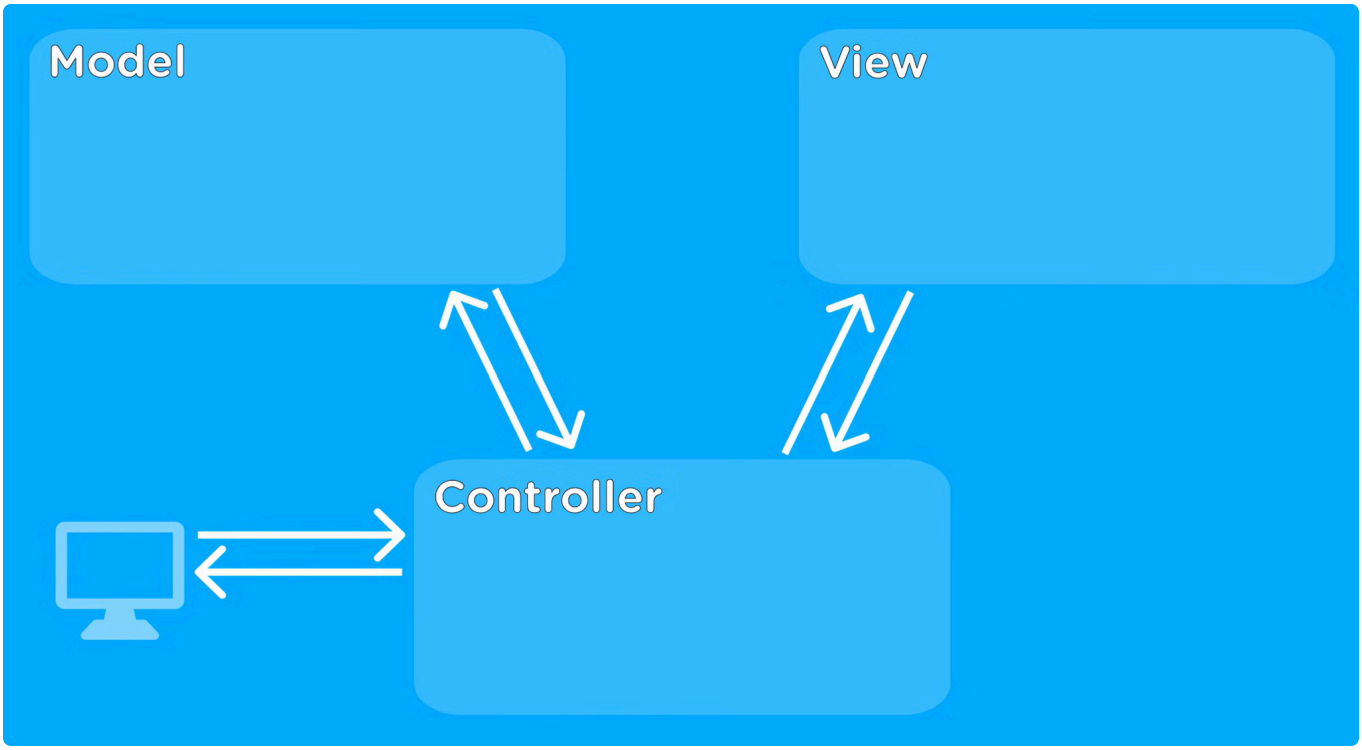
1. Object Oriented Design Patterns

Design patterns make code more readable, easier to make changes to, and less complex. We implemented various design patterns in our project, most notably, Model–View–Controller (MVC), for this exact purpose.

MFBook is the Model that receives data from the Controller, handles data validation, processes data, and returns data. In the context of our project, MFBook validates the timeslot, saves appointments and accounts to .csv, and returns saved appointments and accounts.

The JFrames (GUI) is the View. They are in charge of displaying information, such as appointments, to the user.

the logic of our project is rather simple, so the Controller is also part of the Jframe files. They are implemented through event handling methods in classes like BookingTab and CancelBooking.



In our project, there is also a singleton-like shared state. For each run, only one instance of MFBook is created, guaranteeing that ArrayLists allAppts and students are shared universally.

As our project became more complicated, the merits of these patterns really came to shine. When a new issue appears, we don't need to try to recall where the implementation is. It is very easy to locate the code responsible for that particular feature, as everything is logically organized.

2. Git

Git is a free and open source distributed version control system. For the purpose of versioning and keeping files updated, we created a Github Repo at https://github.com/FayeThePaye/metaforum-booking. We create new branches when implementing new features, and then merge the commit with main.

This made our work process significantly easier, not only for code-sharing, because as Anika and I added parameters of the room, appointment, and student classes one by one as our project progressed, for the sake of ease of testing. Often times a solution would not work; Git is handy in those scenario as it makes it easy to locate a previous version that did work, and revert to it.

Merge pull request #1 from FayeThePaye/date-check

FayeThePaye authored yesterday

Merge branch 'master' into date-check

FayeThePaye authored yesterday

- implemented date checking logic

faye committed yesterday

Commits on Oct 15, 2026

Merge origin/master

au120 authored and au120 committed 5 days ago